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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

PROGRAM: B.Tech

Course Title : Microprocessors and Microcontrollers

Course Code: BECE204L

School: SENSE

Project report

Title:

AI-Powered Railway Health Monitoring and Autonomous
Infrastructure Reporting System

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Introduction & Background:

The railway network is the most important part of global transportation. Every day, it moves millions of people and tons of goods. But environmental factors, thermal expansion, and mechanical wear are always making the structure of railway tracks weaker. Traditional ways of checking things depend a lot on manual labor and regular physical checks, which take a lot of time and can be wrong. This project presents an AI-Powered Railway Health Monitoring System aimed at automating the identification of structural anomalies through real-time sensor data and predictive analytics.

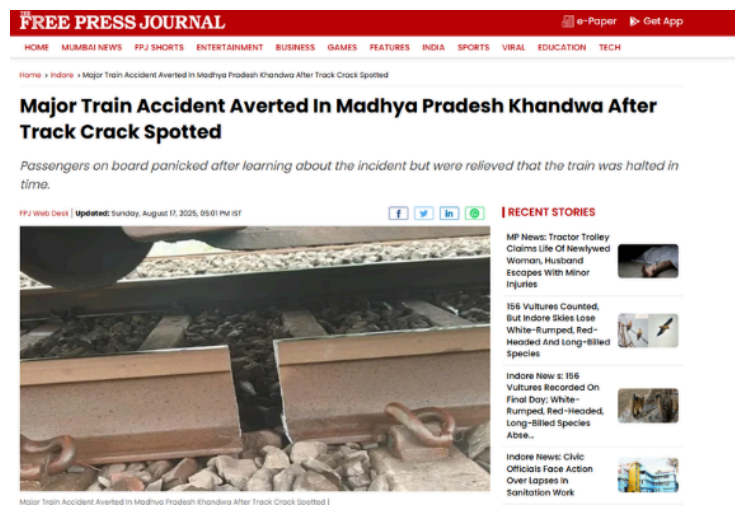


Image 1: Clipping from free press Journal

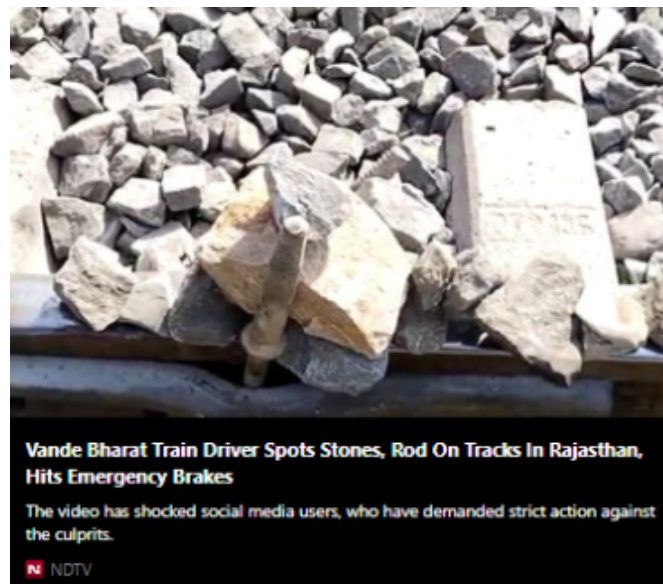


Image 2: Clipping from NDTV news

The motivation behind this invention stems from the increasing frequency of railway derailments caused by undetected track fractures and widening rail joints.

From image 1 and 2 the increasing concern in this field was identified and hence, in order to overcome this issue of oversight or negligence due to human this project is developed.

Aim/Objectives:

AI-Powered IoT System for Real-Time Railway Fault Detection and Autonomous Infrastructure Reporting.

1. **Automating Fault Detection:** Eliminating the need for manual track inspections by continuously monitoring structural vibrations
2. **Dynamic Safety Prioritization:** Implementing a multi-tier safety logic (Safe, Maintenance, Urgent) based on the frequency and severity of detected faults per track section.
3. **Autonomous Incident Reporting:** Streamlining the maintenance workflow by automatically generating timestamped PDF reports containing precise GPS coordinates for rapid repair crew deployment.
4. **Enhancing Infrastructure Longevity:** Reducing the risk of derailments and track failures through predictive analytics and real-time health scoring in intelligent transportation environments.

Block Diagram:

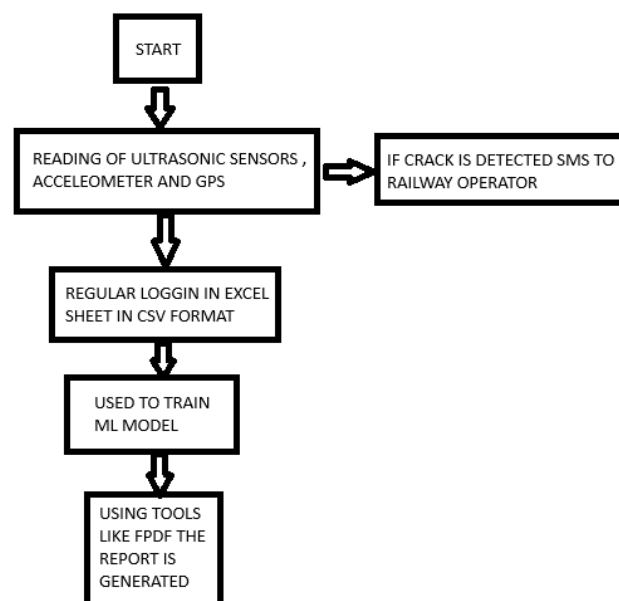


Fig 1: Block diagram

1. Measurements are taken from ultrasonic sensors and accelerometers are regularly taken and stored along with their GPS locations.
2. In case of sudden increase in ultrasonic sensor an alert is generated and sms is sent to railway operator using gsm module
3. The data stored is in format of Distance reading from ultrasonic sensor, vibrations from accelerometer and location (Longitude and Latitude).
4. This data is processed using Rainforest Machine learning model.
5. Data is first processed using the following way.
6. The track is divided into many smaller sections.
7. Then the linear correlation is found in between the vibrations and crack details and then depending on this readings severity of crack is determined.
8. Following which this data is presented using FPDF (Open source extension).

Due to the unavailability of a comprehensive dataset on Kaggle that met our specific feature requirements, synthetic data was programmatically generated using NumPy and Pandas. This approach allowed for the creation of a controlled environment to validate model logic and data pipeline integrity.

Simulation Images:

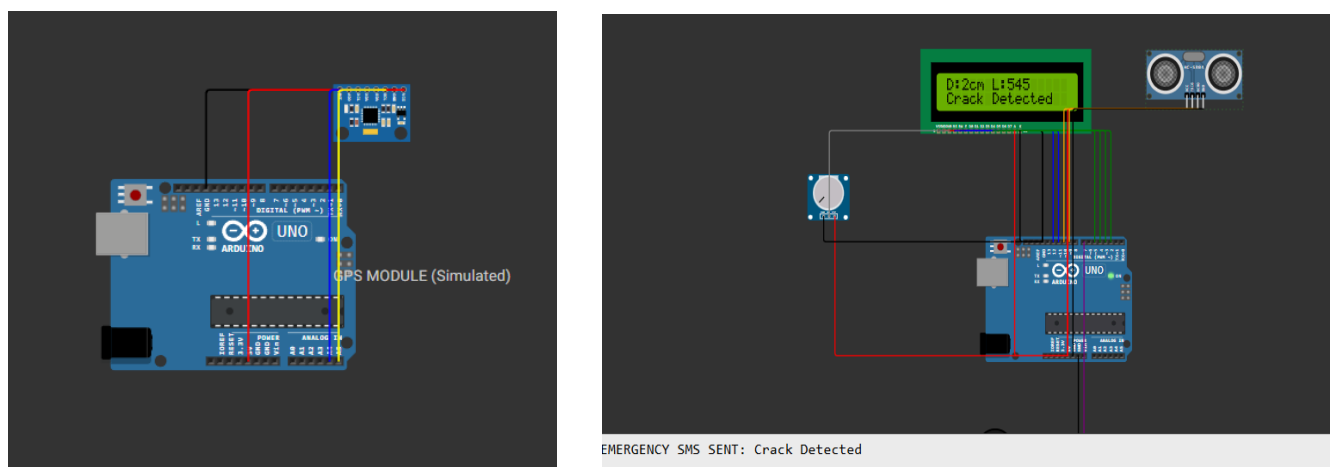


Fig 2 & 3 : Images showing reading from MPU6050 and ultrasonic sensor

Description:

The invention is an automated railway health monitoring and reporting framework that integrates an Arduino microcontroller with sensors. The system is designed to provide real-time structural analysis of railway tracks by capturing data. This data is processed through a localized Machine Learning (ML) engine to determine safety thresholds without manual intervention.

Flowchart/Algorithm:

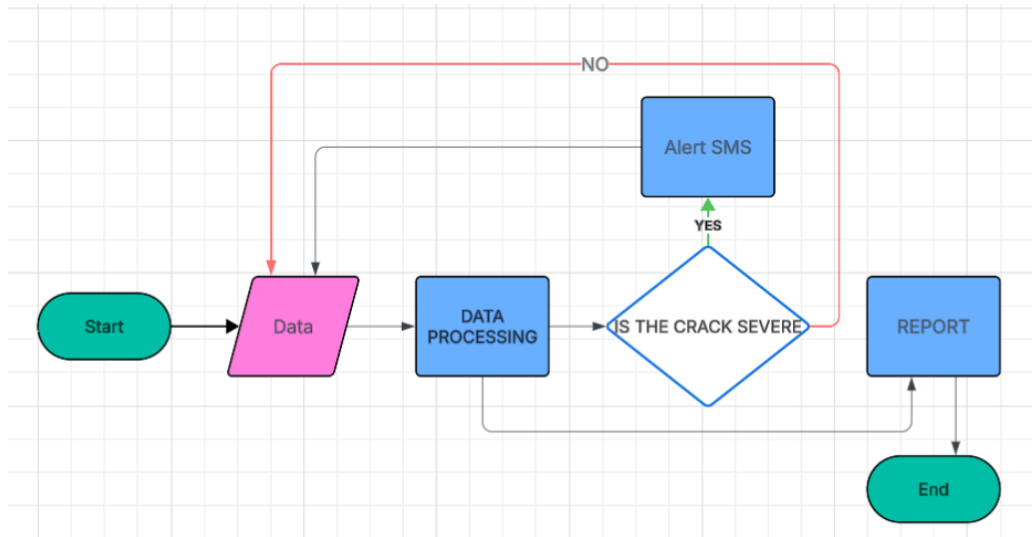


Fig 5 Flowchart

1. Start.
2. Initialize Hardware: Power on the Arduino Mega 2560 and stabilize the sensor array (Ultrasonic Gap sensors and Vibration Transducers).
3. Data Acquisition: Read distance values (mm) from the Gap sensor.
Read mechanical vibration levels (g) from the Accelerometer.
Capture current GPS coordinates for the specific track section.
4. AI Processing: Feed the captured telemetry into the Random Forest Classifier.
5. Fault Classification: The ML model classifies the reading as High, Medium, or Low Priority.
6. Section Aggregation: Count the number of "High Priority" faults within the specific Track Section.
7. Safety Status Determination:
8. If Fault Count ≥ 6 : Set Status to URGENT REPAIR.
9. If Fault Count is between 4 and 5: Set Status to MAINTENANCE.
10. If Fault Count ≤ 3 : Set Status to SAFE.
11. Report Generation:
12. Compile results into a table.
13. Identify the section with the highest priority for the "Top Deployment" header.
14. Format data into a PDF document with a Times New Roman font and a timestamped footer.
15. Output: Save/Print the PDF report and display the target GPS for repair crews.
16. End

Fault detection mechanism:

The Random Forest (RF) model was chosen because it is an "Ensemble Learning" method. This means instead of relying on one single decision-making path, it consults a "forest" of many decision trees and takes the majority vote.

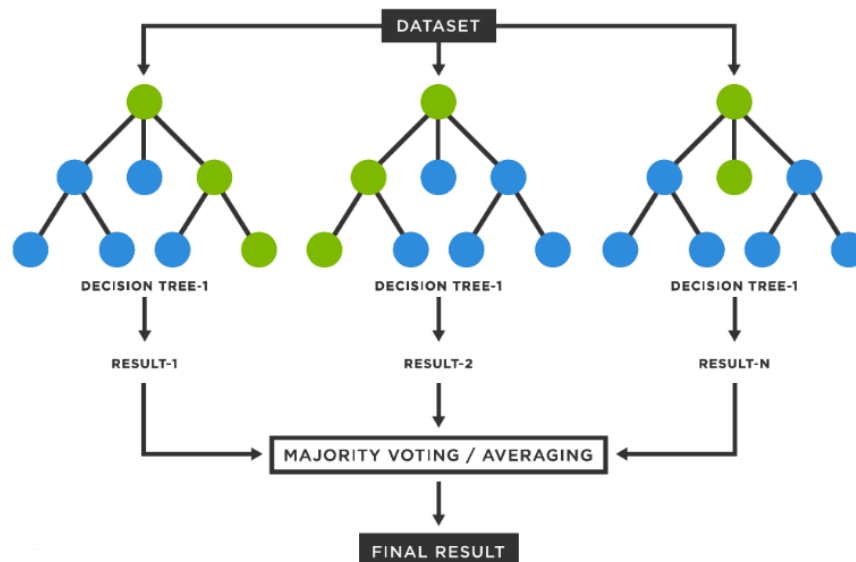


Fig 6 Random forest model Algorithm

Reasons to use Random Forest model

1. Railway environments are "noisy" due to weather, wind, and passing trains. Random Forest is naturally resistant to outliers and noisy data points that might confuse a simpler model.
2. High Performance on Small Datasets:
Since you are likely using a simulated dataset or a laboratory-scale sample size, Random Forest performs significantly better than "Deep Learning" (Neural Networks), which require millions of rows of data to work.
3. Prevents Overfitting:
A single Decision Tree often "memorizes" the training data too well and fails in the real world (overfitting). By averaging many trees, Random Forest ensures that the model generalizes well to new, unseen track sections

Images:

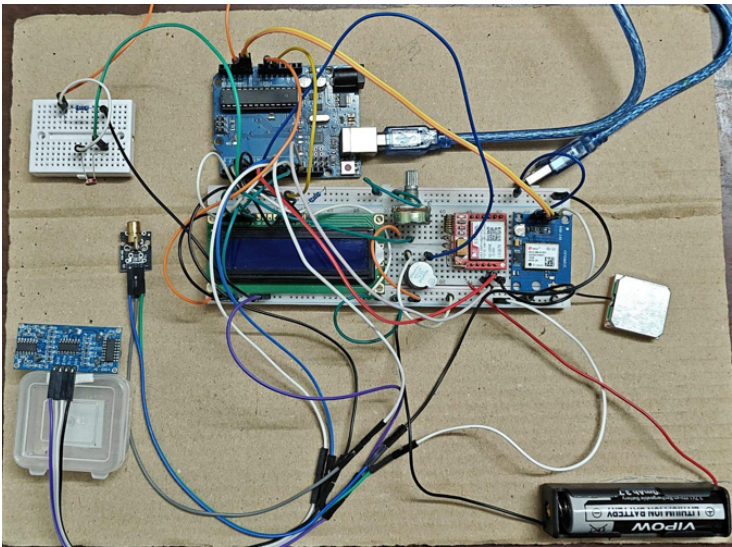


Fig 7 hardware connections

Railway_Times_Report_1603.pdf

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RAILWAY INFRASTRUCTURE HEALTH REPORT

1. CRITICAL DEPLOYMENT TARGET

TARGET: Section SEC_3 (8 Cracks Detected)

GPS COORDINATES: 22.576309001490824, 88.36392671215756

2. EXECUTIVE SUMMARY

Total Sections Analyzed: 100

Urgent Repairs Required (6+ cracks): 39

Maintenance Required (4-5 cracks): 45

Section ID	Crack Count	Safety Status
SEC_3	8	URGENT REPAIR
SEC_7	8	URGENT REPAIR
SEC_44	8	URGENT REPAIR
SEC_42	8	URGENT REPAIR
SEC_35	8	URGENT REPAIR
SEC_49	8	URGENT REPAIR
SEC_97	8	URGENT REPAIR
SEC_5	7	URGENT REPAIR
SEC_33	7	URGENT REPAIR
SEC_27	7	URGENT REPAIR
SEC_71	7	URGENT REPAIR
SEC_26	7	URGENT REPAIR
SEC_36	7	URGENT REPAIR
SEC_40	7	URGENT REPAIR
SEC_46	7	URGENT REPAIR
SEC_60	7	URGENT REPAIR

Fig 8 Infrastructure Report

Literature review:

Prior Patents and Publications:

S.No	Title	Authors	Publication / Patent No.	Key Features	Limitations
1	Efficient model for IoT based railway crack detection system	S. Sakena Benazer, M. Sheik Dawood, Sulochanan Karthick Ramanathan, G. Saranya	10.1016/j.matpr.2020.11.743	It uses RF (Radio Frequency) modules to transfer information at high speeds to a control room or monitoring unit.	· No automated system to decide the severity of crack · No section wise report counting the number of cracks

Novelty/Gap:

1. **Multi-Sensor Fusion for Structural Health:** Unlike single-sensor optical methods, this system utilizes a fusion of ultrasonic gap analysis and mechanical vibration data to detect non-visible structural stressors.
2. **Implementation of Machine Learning for Priority Scoring:** The invention introduces a Random Forest Classifier into the railway monitoring pipeline, enabling the system to "learn" and distinguish between standard operational noise and critical safety risks.
3. **Autonomous Documentation:** The system features an end-to-end automation sequence that converts raw sensor data into a fully formatted PDF Infrastructure Health Report without human intervention.

Future Scope:

Though the current prototype successfully showcases the seamless integration of AI-based fault detection and self-reporting capabilities, further improvements are suggested as follows:

- **Edge Computing Integration:** In upcoming versions, it is possible to switch from centralized processing using AI (NVIDIA Jetson, Raspberry Pi) technology to Edge AI, which will allow the execution of the Random Forest algorithm in real-time without delays of several milliseconds.
- **LWAN Connectivity Integration:** It will become feasible to use the system in remote and rural regions where cellular coverage is weak, thanks to the implementation of LPWAN technology. As a result, the sensor network will have the ability to send infrastructure health information wirelessly, even from up to 15 kilometers away.
- **Detection of Multiple Types of Faults:** Apart from cracks in the rails, it will be possible to detect other faults, such as ballast deterioration, rail deformation caused by excessive temperatures, and obstruction of the tracks due to biological agents.
- **Maintenance Prediction (Digital Twin):** With data collected over many years, the system will become a Digital Twin for the railways. This will make it possible for the authorities to foresee that cracks are about to be formed before they occur, thus shifting from “detection” to “prevention.”

Experimental validation results:

Through integrated hardware simulation and software analysis, the following results were validated:

- **Accurate Structural Fault Classification:** The Random Forest AI model successfully identified "High," "Medium," and "Low"
- **Threshold-Based Safety Logic:** The system correctly categorized track sections (Safe, Maintenance, Urgent) based on the specific crack count thresholds.
- **Dynamic PDF Report Generation:** Verified the autonomous creation of a professional infrastructure report immediately following data processing.

Project Photo:



Project Video Link:

[▶ MPMC_Project](#)